

Mehedi Hasan

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Education		
April 2024 – Feb 2026	Department of Medicine, Bangladesh Agricultural University M.S. in Medicine (CGPA 3.696 out of 4.00) Supervisor: Prof. Dr. Taohidul Islam	
Thesis: Prevalence and antimicrobial resistance pattern of <i>Acinetobacter baumannii</i> from veterinary hospitals and pet animals in Bangladesh		
January, 2018 – February, 2024	Bangladesh Agricultural University Doctor of Veterinary Medicine (DVM) (5 Year) (CGPA: 3.577 out of 4.00)	
Research Experience		
Project 1: Prevalence and antimicrobial resistance pattern of <i>Acinetobacter baumannii</i> from veterinary hospitals and pet animals in Bangladesh (at Dept. of Medicine, Bangladesh Agricultural University)		
Project 2: Assessment of veterinary waste management practices and detection of antibiotic-resistant <i>Klebsiella pneumoniae</i> in veterinary wastes of Bangladesh		
<u>Responsibilities:</u>		
- Conducted extensive field-based sample collection from veterinary hospitals, pet animals, and waste sites across multiple districts, totaling approximately 600 clinical and environmental samples - Performed bacterial isolation, biochemical characterization, and molecular identification of priority pathogens (<i>Acinetobacter baumannii</i> and <i>Klebsiella pneumoniae</i>) using species-specific and multiplex PCR assays - Determined antimicrobial susceptibility profiles using standard disk diffusion/broth microdilution methods and screened isolates for clinically relevant resistance genes (e.g., carbapenemase genes) - Designed and implemented structured surveys and field observations to assess veterinary waste management practices in Bangladesh - Conducted systematic literature reviews, data analysis, and contributed to manuscript preparation for peer-reviewed journals		
Professional Experience		
February 2023– Present	Research Assistant Population Medicine and AMR Laboratory Bangladesh Agricultural University, Bangladesh	
<u>Responsibilities:</u>		
- Microbial detection: Carried out microbiological analysis encompassing cultural isolation, biochemical characterization, and molecular confirmation of key zoonotic and foodborne pathogens (<i>Salmonella</i> spp., <i>S. aureus</i>, <i>L. monocytogenes</i>, <i>E. coli</i>, <i>Streptococcus</i> spp., and <i>Campylobacter</i> spp.) - Assessing antimicrobial resistance: Investigated antimicrobial susceptibility and resistance gene profiles to characterize phenotypic and genotypic AMR patterns in priority bacterial isolates - Biofilm assay: Assessed biofilm-forming capacity using microtiter plate assay, contributing to understanding of virulence and persistence mechanisms in clinical and environmental isolates - Bioinformatic analysis: Conducted phylogenetic reconstruction and amino acid substitution analysis of <i>E. coli</i> and FMDV using MEGA 12 and BioEdit to support evolutionary and epidemiological interpretation of circulating strains - Epidemiological and economic study: Design and execution of a large-scale cross-sectional epidemiological and economic impact study on Lumpy Skin Disease (LSD) in cattle, covering 395 farms across 16 districts of Bangladesh. Estimated attack rates, mortality rates, and farm-level production losses using primary data collection and advanced statistical analysis. - Antimicrobial use quantification: Used UDD (Used Daily Dose) and DDD (Defined Daily Dose) matrices to quantify antimicrobial use in 300 broiler farms across Bangladesh		

7. Statistical analysis: Applied spatial mapping (QGIS) and advanced statistical models including GLM, GLMM, GEE, polynomial regression and Kaplan-Meier survival analysis in SPSS and R to identify risk factors and characterize disease dynamics
8. Manuscript preparation: Contributed to several (close to 6) manuscript preparation
9. Mentoring: Mentoring 4 junior MS fellows and research assistants in laboratory methods, data analysis, and scientific writing

April 2024- **Teaching Assistant**
December 2025 Prof. Dr. Taohidul Islam

Responsibilities:

1. Conducted hands-on QGIS practical classes for first-semester Master's students, including live demonstration of spatial analysis workflows and direct supervision during sessions
2. Assisted in course material development including lecture notes and examination question preparation, and contributed to student performance evaluation
3. Provided informal one-on-one academic support to Master's students, assisting with research methodology, data analysis, and thesis-related queries outside scheduled classes

September 2023- Internship at Veterinary Hospitals and Clinics, Military, Dairy Farms, BRAC Bull and Buck
January 2024 Station, Livestock Research institute, Bangladesh National Zoo, Central Disease Investigation Laboratory (CDIL)

Responsibilities:

1. Performed patient handling and restraint across multiple species including small animals, livestock, poultry, and birds, and assisted in clinical examination, history taking, and disease diagnosis alongside senior clinicians
2. Participated in surgical procedures including spaying, neutering, Caesarean section, fracture management, amputation, hernia repair, and wound management as a surgical assistant
3. Conducted post-mortem examinations of sheep, goat, calf, and chicken, with subsequent sample collection and processing for laboratory investigation

Publication

1. Runa, MA, Tasnuva, S, Rahman, A, Rahman, MM, **Hasan, M.**, et al., (2026), Occurrence, virulence gene profile, and antimicrobial resistance of *Campylobacter* spp. in ready-to-eat and processed chicken and fish products in Mymensingh municipality of Bangladesh, International Journal of Food Science. <https://doi.org/10.1155/ijfo/9923913>
2. **Hasan, M.**, Lira, SA., Islam, A., et al., First report of carbapenem-resistant *Acinetobacter baumannii* in veterinary hospitals and pet animals in Bangladesh. Under Review at *Microbiology Open*
3. Rahman, A., **Hasan, M.**, Khatun, Z., et al., (2026), Epidemiology of recent outbreaks of Foot-and-mouth-disease in cattle and emergence of SEA/Mya-98 topotype in Bangladesh. Under review at *Bulletin of the National Research Center*
4. Jahan, I., **Hasan, M.**, Sharmy, ST. et al., Economic impact and epidemiology of Lumpy Skin Disease in cattle in Bangladesh. Under review at *Preventive Veterinary Medicine*
5. Miti, AA, **Hasan, M.**, Suchy SA, et al, Influence of cow and herd level factors on somatic cell count and total bacterial count in dairy cow milk. Under review at *Journal of Dairy Science*

Manuscript in Preparation:

1. Lira, SA., **Hasan, M.**, Islam, A., et al., Assessment of veterinary waste management practices and detection of antibiotic-resistant *Klebsiella pneumoniae* in veterinary wastes of Bangladesh.
2. Turna AA, Maliha M, **Hasan M.**, et al., Isolation and characterization of biofilm-forming *Salmonella* spp. from poultry drinking water system.
3. Maliha, M., Turna AA, **Hasan M.**, et al., Detection of biofilm-forming methicillin-resistance *staphylococcus aureus* in drinking water systems of layer farms.
4. Ali, SB., Islam, A., Islam, MM, **Hasan, M.**, et al., Rumen health assessment in dairy cows: influence of cow and farm-level factors.

Conference proceedings:

Oral:

1. International Poultry Show and Seminar, World's Poultry Science Association, Bangladesh Branch (WAPSA-BB), 2025

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- Turna AA, Maliha M, **Hasan M**, Rahman A, et al., “Isolation and characterization of biofilm-forming *Salmonella* spp. from poultry drinking water system”.
 - Parvin MS., Maliha M., **Hasan M.**, Rahman A., et al., “Exploring Commensal *E. Coli* Carrying Carbapenems Resistance Genes Isolated from Frozen and Retail Chicken Meat in Bangladesh”
2. 31st BSVER Ascon, Bangladesh Society for Veterinary Education and Research
 - Rahman MM, **Hasan M**, Keya AN, et al., (2025), “Assessment of vaccination, biosecurity practices, and the use of alternatives to antibiotics on small-scale broiler farms in three districts of Bangladesh”.

Poster:

1. World Small Animal Veterinary Association Congress, in Warsaw, Poland (13 – 15 October,2026)
Hasan, M., Lira, SA., Islam, A., et al., (2026), First report of carbapenem-resistant *Acinetobacter baumannii* in small animal hospitals and pet animals in Bangladesh (Accepted for E-poster)
2. World Veterinary Poultry Association Congress, 2025 in Sarawak, Malaysia
 - Maliha, M., Turna AA, **Hasan M**, et al., (2025), “Detection of Biofilm-Forming Methicillin-Resistance *Staphylococcus aureus* in Drinking Water Systems of Layer Farms”.

Academic service:

1. **Peer Reviewer:** *Tropical Animal Health and Production*, 2026
2. Member, Bangladesh Veterinary Council
3. Community Scientific Outreach:
 - a. Conducted antimicrobial resistance (AMR) awareness sessions among rural dairy farmers, promoting responsible antibiotic use and One Health practices
 - b. Participated in livestock vaccination campaigns supporting disease prevention initiatives

International exposure

1. Participant, Sakura Science Exchange Program, Shimane University, Japan (2025)
 - a. Awarded Best Presenter,
 - b. Training in advanced microscopy, histology, and laboratory animal handling
2. Participated in Hult Prize, Regional (South Asia) round. 2024

Conferences

1. International Poultry Show and Seminar, 2025, Organised by World Poultry Science Association - Bangladesh Branch
2. 30th BSVER (Bangladesh Society for Veterinary Education and Research) conference, 2024
3. CEVET BD (Continuing Education for Vets in Bangladesh) conference, 2020

Honor and awards

1. National Science and Technology (NST) Fellowship awarded by Ministry of Science and Technology, Bangladesh
2. Best presenter at Sakura Science Exchange Programme (2025), Organised by Department of Neuroscience, Shimane University, Japan.

Language proficiency

1. English- IELTS: 7.5 (Reading: 9, Listening: 7.5, Writing: 6.5, Speaking: 6.5)

Voluntary activities

1. Student Representative, Social Harmony Committee, Mymensingh, Bangladesh, (March 2021 –2024).
Represented student body in efforts to combat communal and religious violence and promote interfaith harmony.
 2. Joint Secretary, Hashimukh (Voluntary Organisation), Bangladesh Agricultural University, (January, 2018–December 2022).
Organized flood relief, public health awareness, and educational outreach programs
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